

	Silver	Gold	Copper	Nickel
Conductivity (MS/m)	Green	Yellow	Light green	Red
Resistivity (nΩm)	Green	Yellow	Light green	Red
Density (g/cm ³)	Yellow	Red	Light green	Green
Cost (\$/kg)	Yellow	Red	Light green	Green
Oxidation resistance	Light green	Green	Red	Yellow

Fig. 3: Colour-coded comparison chart (Green: favourable, Light green: slightly favourable, yellow: moderate, red: unfavourable) (Credit: IDTechEx)

	Epoxy	Acrylics	Polyimides	Silicone
Modulus	Green	Green	Yellow	Red
Adhesion Strength	Green	Light green	Yellow	Red
Cure Time	Light green	Green	Yellow	Red
Cure Temperature	Yellow	Green	Red	Light green
Functional Temperature	Light green	Red	Green	Yellow

Fig. 4: Colour-coded comparison chart (Green: favourable, Light green: slightly favourable, yellow: moderate, red: unfavourable) (Credit: IDTechEx)

with ECAs.

4. Lack of self-alignment. Although solders allow perfect component placement, even when they are placed incorrectly, with ECAs more precision is required.

Applications

Consumer electronics. ECAs are perfect for placing a surface mount device. Since more electronics goods are expected to be produced each year, ECAs have a lot of potential to grow in this large market. However, the speed with which they penetrate may not be fast.

Currently, they share the market with other widely used solders. But this may change as they are seen to be a much more superior option for certain applications. Due to the high cost, they are unlikely to replace solders soon in the consumer electronics market, which uses only ICAs as of now.

The key things to aim for when choosing or developing an ECA are low cost and easy processability. High cost and brittle connections are

features to avoid as too many of them cause problems. Consumer electronics are high-volume goods and hence it is best to make most connections as cheap and quick as possible.

Automotive electronics. Being a highly established market, there is a lot of room for ECAs to grow as car electronic systems are becoming more advanced with an aim to increase sales by 2035. Currently, almost all ECAs used in automotive electronics are ICAs. ACAs have better miniaturisation capabilities than ICAs but are forecast to grow at a very slow rate.

Desirable properties are easy processability and high chemical resistance. Features to avoid are brittle connections and low reliability. Cars go through a lot of vibrations and stress in their daily use, needing highly reliable connections that can work overtime. Thus, high chemical resistance to withstand rugged environments and easy processability are beneficial for large-scale production of car electronics.

Display technology. The market

for ECAs in display technologies is expected to grow at a steady rate, although not as rapid as other technologies. ACAs and, more specifically, ACFs are the ECAs that are used in displays. There is not much potential for ICAs in this market. Desirable properties are low temperature, rapid curing, fine pitch capacity, and thin film thickness. Poor adhesion can be a critical failure point for display applications because of which peeling from the screen can occur.

Some other applications include EMI shielding, RFID tags, photo-voltaics, and aerospace. Printed electronics, in-mould electronics, and wearable electronics are those emerging markets where ECAs hold a lot of promise.

ECAs, the successor to solder?

After getting a glimpse into what ECAs are all about, and analysing their benefits and shortcomings, it does seem that they are set to become a key conductive joining technology. A prediction of their application in emerging markets is seen, but with the rapid advancement in electronics technology, such as miniaturisation of components, they can be conveniently used for connecting circuitry across a wide range of uses, providing joins that are mechanically strong and electrically conductive. **EFY**

This article has been prepared by Vinay Prabhakar Minj, Technology Journalist at EFY, based on webinar titled 'Electrically Conductive Adhesives (ECAs): The Successor to Solder?' organised by IDTechEx.

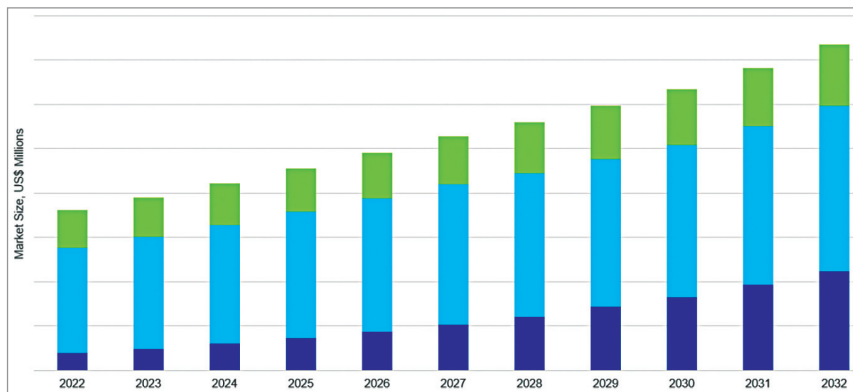


Fig. 5: Global ECA market forecast by type of ECA, 2022-2032 (Credit: IDTechEx)